

Factorising

Example

Factorise: $6a + 12$

Factorising is the reverse of expanding brackets.
We need to put our expression back into brackets.

Step 1:

We need to identify the common factors between both terms. Basically, what factors are common to $6a$ and 12 ? These could be numbers, letters, or both.

Common factors: 1, 2, 3, 6

Step 2:

Select the highest common factor and place this on the outside of the brackets.

Common factors: 1, 2, 3, 6

$6(\quad)$

Step 3:

To fill the inside of the brackets, you need to ask yourself: "What do I need to multiply 6 by to obtain the expression I started with?"

In other words:

- 6 multiplied by something is equal to $6a$.
- 6 multiplied by something is equal to 12.

$6(a + 2)$

Your final answer is $6(a + 2)$

Example

Factorise: $4x^2 + 8x$

Step 1:

We need to identify the common factors between both terms. Remember, these could be numbers, letters, or both.

Common factors: x , 2, 4

Step 2:

Select the highest common factors and place these on the outside of the brackets.

Common factors: x , 2, 4 - this time we will have a letter and a number.

$4x(\quad)$

Step 3:

To fill the inside of the brackets, you need to ask yourself: "What do I need to multiply $4x$ by to obtain the expression I started with?"

In other words:

- $4x$ multiplied by something is equal to $4x^2$.
- $4x$ multiplied by something is equal to $8x$.

$4x(x + 2)$

Your final answer is $4x(x + 2)$